Final exam, Math 170S, Winter 2018
Instructor: Elizaveta Rebrova

Printed name: SOLUTIONS

Signed name: ______

Student ID number: ______

Instructions:

- Read problems very carefully. If you have any questions please ask.
- **The correct final answer alone is not sufficient for full credit - you should explain your solutions,** saving enough time to attempt all the problems.
- You final answers does not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- Books, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	8	
4	12	
5	10	
6	10	
7	10	
8	10	
9	10	
Total:	90	

1. ***For this problem only:*** *give a specific, but short (one sentence) answer. If several things can be done with the help of a specific test, it is enough to clearly identify one of them.* ***Explain all the notations you use.***

   (a) (2 points) What hypothesis can be tested with the help of Wilcoxon test?

   That median of the distribution of $X = m_0$

   (b) (2 points) What hypothesis can be tested with the help of ANOVA test(s)?

   That some factor A has no effect on the distribution of X

   (c) (2 points) In the model for simple linear regression we assume that data points (x_i, Y_i) are close to a line, namely, $Y_i = \alpha_1 + \beta x_i + \varepsilon_i$. What assumption do we make about the noise ε_i?

   $\varepsilon_i \sim N(0, \sigma^2)$ and independent

   (d) (2 points) If we plot sample quantiles of some data versus quantiles of standard normal distribution (q-q plot) and the resulting points are close to a straight line, what conclusion can we make?

   That data is approximately normally distributed

   (e) (2 points) Suppose we are using some test for a hypothesis, and we would like to improve both Type I and Type II errors with the same test. What can we do?

   Increase n (sample size)

Δ

2. Consider the distributions $N(\mu_X, 400)$ and $N(\mu_Y, 225)$. Let $\theta = \mu_X - \mu_Y$. Let $\bar{x}$ and $\bar{y}$ denote the observed means of two independent random samples, each of size n, from the respective distributions. Say we reject $H_0 : \theta = 0$ and accept $H_1 : \theta > 0$ if $\bar{x} - \bar{y} \geq c$. Let $K(\theta)$ be the power function of the test.

(a) (2 points) By definition, power function

$K(t) = \mathbb{P}($ to reject Ho if $\theta = t$ $)$.

(b) (1 point) The distribution of

$\bar{x} - \bar{y} \sim N\left(\theta, \frac{400 + 225}{n}\right) = N\left(\theta, \frac{625}{n}\right)$

Give the parameters of the distribution too.

(c) (7 points) Find n and c, so that $K(0) = 0.05$ and $K(10) = 0.90$, approximately.

$$\begin{cases} P(\bar{X} - \bar{Y} \geq c \mid \theta = 0) = 0.05 & (1) \\ P(\bar{X} - \bar{Y} \geq c \mid \theta = 10) = 0.9 \end{cases}$$

(1) $P\left(\frac{\bar{X} - \bar{Y} - \theta}{25/\sqrt{n}} \geq \frac{c - \theta}{25/\sqrt{n}}\right) \sim P\left(Z \geq \frac{c\sqrt{n}}{25}\right) = 0.05$

$\frac{c\sqrt{n}}{25} = 1.64$

(2) $P\left(\frac{\bar{X} - \bar{Y} - \theta}{25/\sqrt{n}} \geq \frac{c - 10}{25/\sqrt{n}}\right) \sim P\left(Z \geq \frac{(c-10)\sqrt{n}}{25}\right) = 0.9$

$\Rightarrow \frac{(c-10)\sqrt{n}}{25} = -1 \cdot 1.28$

$$\begin{cases} c\sqrt{n} = 1.64 \cdot 25 \\ (c-10)\sqrt{n} = -1.28 \cdot 25 \end{cases} \Rightarrow 10\sqrt{n} = 25(2.92)$$

$\boxed{c = \frac{1.64 \cdot 25}{7.5}}$

$n = 7.5^2 \approx 56$ ≈ 75

3. Let $X_1, X_2, ..., X_{50}$ be a random sample from Poisson distribution with parameter $\lambda > 0$.

(a) (4 points) We will reject H_0: $\lambda = 2$ and accept H_1: $\lambda > 2$ if the observed sum $\sum_{i=1}^{50} X_i$ is bigger than 110. Write a numerical expression that computes the exact Type I error probability.

(b) (4 points) Give a normal approximation for the probability obtained in part (a). In this part you should get a numerical answer.

Hint: The Poison distribution with parameter λ has pmf $p_X(k) = e^{-\lambda}\frac{\lambda^k}{k!}$ for $k = 0, 1, 2, \ldots$. Also, $\mathbb{E}(X) = \mathrm{Var}(X) = \lambda$. Also, if $X_i \sim Poi(\lambda)$ and independent, then $\sum_{i=1}^{m} X_i \sim Poi(m\lambda)$

(a) $P(\text{reject } H_0 \mid H_0) =$

$$= P\Big(\underbrace{\sum_{i=1}^{50} X_i}_{\substack{\shortparallel \\ Poi(50\cdot 2)}} > 110 \,\Big|\, \lambda = 2\Big) = P(Poi(100) > 110) = \sum_{k=111}^{\infty} P(Poi(100) = k) =$$

$$= \sum_{k=111}^{\infty} e^{-100}\frac{100^k}{k!} = 1 - \sum_{k=0}^{110} e^{-100}\frac{100^k}{k!}.$$

(b) $P(P > 110)$ $\quad P \sim Poi(100)$ $\quad P \approx N(100, 100)$

$\frac{P-100}{10} \approx N(0,1)$

$$= P\left(\frac{P-100}{10} > \frac{110-100}{10}\right) = P(Z > 1) \approx 0.16$$

$\boxed{d}$

4. (~~10~~ 8 points) Let $X_1, \ldots, X_{10}$ be a random sample from $N(0, \sigma^2)$.

✓ (a) (~~5~~ 4 points) Show that most powerful critical region for testing $H_0 : \sigma^2 = 9$ against $H_1 : \sigma^2 = 2$ can be defined using statistic $\sum_{i=1}^{10} x_i^2$.

(b) (~~5~~ 4 points) Find the best critical region of the size $\alpha = 0.05$. For the test defined by this critical region find probability of Type II error.

Hint: Recall that $\sum_{i=1}^{n} X_i^2/\sigma^2 \sim \chi^2(n)$.

(a) By Neyman – Pearson Lemma it is enough to find k, such that

$$\frac{L(x_1 \ldots x_n, 9)}{L(x_1 \ldots x_n, 4)} \leq k \quad \text{for } (x_1 \ldots x_n) \in C$$

$$\|$$

$$\frac{\left(\frac{1}{\sqrt{2\pi \cdot 9}}\right)^n \cdot \exp\left(-\frac{\sum x_i^2}{18}\right)}{\left(\frac{1}{\sqrt{2\pi \cdot 2}}\right)^n \cdot \exp\left(-\frac{\sum x_i^2}{2}\right)} = \left(\frac{\sqrt{2}}{3}\right)^n \cdot \exp\left(-\frac{\sum x_i^2}{18} + \frac{\sum x_i^2}{2}\right)$$

$$= \left(\frac{\sqrt{2}}{3}\right)^n \cdot \exp\left(+\left(\sum x_i^2\right)\left(\frac{4}{9}\right)\right) \leq k$$

$$k\left(\frac{3}{\sqrt{2}}\right)^n \geq \exp\left(+\left(\sum x_i^2\right) \cdot \frac{4}{9}\right)$$

$$\ln\left(\frac{3}{\sqrt{2}}\right) n + \ln k \geq +\left(\sum x_i^2\right) \cdot \frac{4}{9}$$

$$\cancel{\sum x_i^2 \leq \frac{9}{4}\left(n \ln\left(\frac{3}{\sqrt{2}}\right) + \ln k\right)}$$

$$\sum x_i^2 \leq \frac{9}{4}\left(n \ln \frac{3}{\sqrt{2}} + \ln k\right)$$

$$C = \{\sum x_i^2 \leq c\}$$

5. (10 points) On some magical island, there were elves, dwarfs, orcs and goblins. A researcher arrives to the island and decides to interview all four spices about their history and traditions. She invites 10 representatives of each folk for an interview. However, she soon finds out that at least some of the island inhabitants are quite mean and aggressive and don't interview well (whereas the rest are quite nice and friendly). So, the researcher wonders: are there more and less aggressive spices on the island? She records the data she got so far with in the following table:

	nice	not nice
elves	3	7
dwarves	5	5
orcs	4	6
goblins	8	2

How can she test on the significance level 10% a hypothesis that being nice is a spices independent feature?

Specify what test you will use. Show all the steps. Make a conclusion (whether the researcher should accept or reject her hypothesis about independence).

We can use χ^2 test for independence of 2 factors

$k=2 \quad h=4 \qquad (k-1)(h-1)=3$

$$Q_3 = \sum_{j=1}^{h}\sum_{i=1}^{k} \frac{\left(y_{ij} - n\cdot\frac{y_{i\cdot}}{n}\cdot\frac{y_{\cdot j}}{n}\right)^2}{n\cdot\frac{y_{i\cdot}}{n}\cdot\frac{y_{\cdot j}}{n}}$$

$$n\cdot\frac{y_{i\cdot}}{n}\cdot\frac{y_{\cdot j}}{n} = \frac{10}{40}\cdot\frac{20}{40}\cdot 40 = 5$$

for $\forall i,j$

So,

$$Q_3 = \frac{(3-5)^2}{5} + \frac{(7-5)^2}{5} + 0 + 0 + \frac{(4-5)^2}{5} + \frac{(6-5)^2}{5} + \frac{(8-5)^2}{5} + \frac{(2-5)^2}{5}$$

$$= \frac{9+9+1+1+4+4}{5} = \frac{28}{5} = 5\frac{3}{5}$$

$\chi^2_{0.1}(3) = 6.25$

$Q_3 < \chi^2_{0.1,3} \Rightarrow H_0$ is accepted

$\boxed{d}$

6. Density function of X is defined as

$$f(t) = \begin{cases} \frac{4}{\theta^2}t, & \text{when } 0 < t < \theta/2, \\ -\frac{4}{\theta^2}t + \frac{4}{\theta}, & \text{when } \theta/2 < t < \theta, \\ 0, & \text{otherwise.} \end{cases}$$

Unknown parameter $\theta \in (0, 2]$.

(a) (5 points) Find estimator of θ by the method of moments.

(b) (3 points) Is it biased or not?

(c) (2 points) For the following observations, give a point estimate for θ:

0.3 0.2 0.5 0.3 0.2 0.2

(a) $\mathbb{E}X = \overline{X} \to$ method of moments estimator

$$\mathbb{E}X = \int_0^{\theta/2} \frac{4}{\theta^2} t \cdot t \, dt + \int_{\theta/2}^{\theta} \left(-\frac{4}{\theta^2} t + \frac{4}{\theta}\right) t \, dt =$$

$$= \frac{4}{\theta^2} \frac{t^3}{3}\Big|_0^{\theta/2} + \left(-\frac{4}{\theta^2}\right) \frac{t^3}{3}\Big|_{\theta/2}^{\theta} + \left(\frac{4}{\theta}\right) \frac{t^2}{2}\Big|_{\theta/2}^{\theta} =$$

$$= \frac{4}{\theta^2} \frac{\theta^3}{8 \cdot 3} - 0 - \frac{4}{\theta^2} \frac{\theta^3}{3} + \frac{4}{\theta^2} \frac{\theta^3}{8 \cdot 3} + \frac{4}{\theta} \frac{\theta^2}{2} - \frac{4}{\theta} \frac{\theta^2}{8}$$

$$= \frac{\theta}{2 \cdot 3} - \frac{4\theta}{3} + \frac{\theta}{2 \cdot 3} + \frac{4\theta}{2} - \frac{\theta}{2}$$

$$= \frac{\theta}{6} - \frac{4}{3}\theta + \frac{\theta}{6} + 2\theta - \frac{\theta}{2} = \frac{\theta}{3} - \frac{4}{3}\theta + 2\theta - \frac{\theta}{2}$$

$$= \theta - \frac{\theta}{2} = \frac{\theta}{2}$$

$$\frac{\theta}{2} = \overline{X} \Rightarrow \boxed{\hat{\theta} = 2\overline{X}}$$

7. (10 points) Suppose that the observation X has exponential density function $f_X(t) = \theta e^{-\theta t}$ for $t \geq 0$, where θ is an unknown parameter. You know that θ is either 2 or 4. Prior the experiment, you have no idea which one is more likely (so, we assume that two options are equally likely). Then you observe in 3 experiments that $X = 7, 2, 1$. Find the posterior distribution of Θ and MAP estimator of Θ. Give numerical values as an answer.

Hint 1: Posterior distribution (as well as prior) is discrete and can take only two distinct values. So, posterior distribution = new probabilities of each possible value of θ.

Hint 2: $e^{20} \gg 8$.

joint density for $X_1, X_2, X_3 = 7, 2, 1$

$$\mathbb{P}(\theta \mid x) = \frac{\mathbb{P}(x \mid \theta) \cdot \mathbb{P}(\theta)}{\sum_{\theta' = 2,4} \mathbb{P}(x \mid \theta') \cdot \mathbb{P}(\theta')} =$$

$$= \frac{\theta^3 e^{-\theta(7+2+1)} \frac{1}{2}}{\underset{\theta=2}{8 \cdot e^{-2 \cdot 10}} \cdot \frac{1}{2} + 4^3 \cdot e^{-4 \cdot 10} \frac{1}{2}}$$

$$\begin{cases} \mathbb{P}(\theta = 2 \mid X_1, X_2, X_3) = \dfrac{4e^{-20}}{4e^{-20} + 32e^{-40}} \\ \mathbb{P}(\theta = 4 \mid X_1, X_2, X_3) = \dfrac{32e^{-40}}{4e^{-20} + 32e^{-40}} \end{cases}$$

Posterior

MAP : $\hat{\theta}$ that maximizes posterior

since $e^{20} > 8$

$32e^{-40} < 4e^{-20}$

$\Rightarrow$ $\underline{\hat{\theta} = 2}$

8. A sample size of $n = 100$ is taken from the production of light switches. A quality control engineer discovered that 10 switches does not meet the quality standard.

(a) (5 points) Suppose that another 500 switches are to be produces using the same technology. What are minimal and maximal amounts of defective switches should we expect with 95% confidence?

Hint: Use confidence interval for the defect rate of the production (=proportion of bad switches) to answer this question.

Confidence interval for p - proportion of defective switches

$$p \in \left[\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right] = \left[0.1 \pm z_{0.025}\sqrt{\frac{0.1 \cdot 0.9}{100}}\right]$$

$$= \left[0.1 - 1.96 \cdot \frac{3}{100};\ 0.1 + 1.96 \cdot \frac{3}{100}\right] \approx [0.1 - 0.0588;\ 0.1 + 0.0588]$$

$$= [0.0422;\ 0.1588]$$

So, from 500 switches, there are at most $500 \cdot 0.1588 \approx \underline{80}$ and at least $500 \cdot 0.0422 \approx \underline{21}$ bad switches

79, 22 are also accepted!

(b) (5 points) Estimate the confidence (=probability) that among the next 500 switches will be at most 60 defective switches.

prior estimate $\boxed{\hat{p} = 0.1}$

$P(\text{at most } \frac{60}{500} \text{ defective rate}) =$

$$P\left(\frac{y}{n} \leq \frac{60}{500}\right) = P\left(\frac{y/n - \hat{p}}{\sqrt{\hat{p}(1-\hat{p})/n}} \leq \frac{6/50 - \hat{p}}{\sqrt{\hat{p}(1-\hat{p})/n}}\right)$$

$$= P\left(Z \leq \frac{\frac{6}{50} - \frac{1}{10}}{\sqrt{0.1 \cdot 0.9/100}}\right) = P\left(Z \leq \frac{1}{50} \cdot \frac{100}{3}\right)$$

$$= P\left(Z \leq \frac{2}{3}\right) \approx \underline{0.74}$$

Δ

9. You test a random number generator which supposed to give 0, 1 or 2 equally likely. You observe 80 zeros, 105 ones, and 115 twos, using the same generator many times in a row.

(a) (5 points) Using the Chi-Square test, do you accept the hypothesis that your generator is good (gives 0, 1 and 2 equally likely indeed) at the 10% significance level? What about 1% significance level?

(b) (3 points) Approximate an interval for the p-value for this test.

(c) (2 points) What is the largest significance level on which you would accept the above hypothesis about fair generator?

(a) 3 classes (0, 1, 2)

$n = 300 = 80 + 105 + 115$

$H_0: p_{10} = p_{20} = p_{30} = \frac{1}{3}$

$$q_{k-1} = q_2 = \frac{(80-100)^2}{100} + \frac{(105-100)^2}{100} + \frac{(115-100)^2}{100} = \frac{20^2 + 15^2 + 5^2}{100}$$

$$= \frac{225 + 400 + 25}{100} = \frac{650}{100}$$

$\{q_2 > \chi^2_{0.1}(2)\} = \{6.5 > 5.99\}$

True $\Rightarrow H_0$ is rejected on 10%

$\{q_2 > \chi^2_{0.01}(2)\} = \{6.5 > 9.2\}$

Not true $\Rightarrow H_0$ is accepted on 1%

(b) $p\text{-value} = P\{\chi^2(2) > q_2\} = P(\chi^2(2) > 6.5) \approx 0.03$

(c) $\alpha = 3\%$ (same as p-value)